Leonardo Serodio

lserodio@andrew.cmu.edu | 978-856-9014 | github.com/leoserodio

EDUCATION

Carnegie Mellon University, Pittsburgh, PA Graduate May 2028
Bachelor of Science in Electrical and Computer Engineering — GPA 3.6/4.0
Groton School, Groton, MA Graduated June 2024

SKILLS

- Software: Python, C, C++, MATLAB; Visual Studio 2022, VS Code, Git, GitHub; Windows, Linux
- Hardware: Verilog, SystemVerilog, VHDL (synthesis and testbench development), FPGA development (AMD/Xilinx Artix-7), Cadence Allegro, signal integrity analysis (router PCB level)
- Networking: Ethernet, Wi-Fi, TCP/IP, IP routing and addressing, MPLS, Nokia SR OS (SROS), Cisco IOS configuration

WORK EXPERIENCE

Nokia — Hardware Development Intern June 2026 – August 2026
- Used VHDL for FPGA applications supporting prototype Nokia network switches.
- Probed and debugged FPGAs directly on the board during bring-up, isolating logic and timing issues on prototype hardware.
- Used Cadence Allegro alongside 20 GHz oscilloscopes and interface testers to troubleshoot power, control, and high-speed interface issues on next-generation routing platforms.

Nokia — Hardware Intern, Regression Lab May 2025 – August 2025
- Supported networking testbeds in lab and data-center environments, configuring and debugging routers, servers, and traffic generators.
- Performed signal-integrity characterization and de-embedding measurements on 200G GAUI-PAM4 interfaces for a next-generation Nokia router using a Keysight UXR-series oscilloscope prototype.

RESEARCH & TEACHING EXPERIENCE

Carnegie Mellon University — Undergraduate Researcher, Pagliarini Lab Summer–Fall 2026
- Building an automated testbed for power side-channel analysis on ASIC chips performing encryption/decryption — using an FPGA/microcontroller to trigger crypto operations and synchronize oscilloscope capture, then automating trace collection and offline correlation power analysis (CPA) in Python to recover secret keys.

Carnegie Mellon University — Teaching Assistant, 18-290 Signals and Systems Fall 2026
- Supporting instruction for the ECE core course covering continuous- and discrete-time signal processing, including LTI systems, convolution, filtering, sampling, and the Fourier and fast Fourier transforms.

PERSONAL PROJECTS

RISC240-ML — Vector/ML-Extended RISC240 Processor
Languages: SystemVerilog, Python
GitHub: github.com/Leo-Personal-Projects/RISC240-ML-Extended
- Extended a SystemVerilog RISC processor with SIMD/vector execution hardware by redesigning the datapath, control FSM, and instruction set to support ML-oriented vector operations.
- Added new hardware modules to the CPU — vector register file, vector ALU, dot-product unit, and accumulator — plus a simple custom assembler and simulator; verified the RTL with an automated Synopsys VCS regression framework (18 tests).

FPGA UDP Traffic Generator — Nexys A7 (Xilinx Artix-7)
Languages: Verilog, SystemVerilog
GitHub: github.com/Leo-Personal-Projects/UDP-traffic-generator
- Designed and implemented a hardware UDP traffic generator on a Xilinx Artix-7 FPGA by integrating an Ethernet MAC with an open-source UDP/IP stack.
- Implemented an application-layer FSM in SystemVerilog to generate, sequence, and transmit UDP packets via AXI-Stream; verified live transmission over Ethernet with Wireshark and a SystemVerilog testbench.

CMU PROJECTS

Zorgian Mind — FPGA Digital Systems Game
18-240 Structure and Design of Digital Systems | Languages: SystemVerilog
GitHub: github.com/Leo-CMU-Projects/Lab4_18240
- Designed and implemented a hardware version of a Mastermind-style guessing game in SystemVerilog, synthesized and deployed on a Xilinx Spartan-7 FPGA (Boolean Board).
- Built a hierarchical RTL design using FSMs, datapaths, and standard components to manage game logic (coin acceptance, pattern loading, Znarly/Zood guess grading, round tracking), with thorough unit/system testbenches and an HDMI display interface for real-time game state.

CERTIFICATIONS

NVIDIA-Certified Associate: Generative AI LLMs (NCA-GenL)
- Industry-recognized NVIDIA credential validating skills in building, integrating, and maintaining applications powered by generative AI and large language models on NVIDIA platforms.